

Curriculum Vitae

Official Address

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h-index: 9 (Scopus), i10 index:8 (google scholar)
Scopus ID: 8846514800

PROFESSIONAL EXPERIENCE

ASSOCIATE PROFESSOR, Department of Biomedical Engineering, Indian Institute of Technology, Hyderabad (2019 June - till date)

ASSISTANT PROFESSOR, Department of Biomedical Engineering, Indian Institute of Technology, Hyderabad (2012 Dec - 2019 June)

Research Area: Lab on chip microfluidics, Computational Biomechanics

PROFESSIONAL EXPERIENCE

(2006 – 2012)

Post Doctoral Research: Nanyang Technological University, Singapore, National University of Singapore

Research Area: Lab on Chip Microfluidic designs for bioanalytical applications

ACADEMIC QUALIFICATIONS

PhD: Nanyang Technological University, Singapore 2002 - 2006
Advisor: Prof. Yang Chun, Charles
Thesis: **Transport and Deposition of Colloidal Particles in Microchannel Flow**

M.E: National Institute of Technology (REC), Trichy, India 1999 – 2001
Specialization: Fluid Mechanics, Heat Transfer, Machine Design

B.Tech: Calicut University, Kerala, India 1994 - 1998
Major: Mechanical Engineering Class: Distinction (Honors)

ACADEMIC EXPERIENCE: (2012 DEC - TILL DATE)

- Independently developed **Biomicrofluidics and Biomechanics Laboratory (BML)** that focuses on the design and fabrication of chip scale microfluidic devices for bioanalytical applications. Some key projects are:
 - Characterization, Trapping and Concentration of Cancer Cells using Dielectrophoresis based Microfluidic devices
 - Paper based Microfluidics (muPADs) for identification of osteoporosis biomarkers
 - Microfluidic immunoassays for Tau protein aggregation - based neurodegeneration in Alzheimer's disease and drug screening
 - LED based optowetting platforms for microfluidic droplet mixing
 - Organ-on-chip, biomimicking microfluidic models for disease modeling
 - Cost-effective In-vitro fertilization (IVF) – on – chip models for sperm selection and embryo culture
- Independently developed computational biomechanics and mechanobiology lab (CBM – part of BML) that focuses on the modeling of biomechanical problems using multiscale approach. The lab houses high end workstations for GPU-intensive computing and data visualization. Some key projects are:
 - Modeling of traumatic brain injury and investigation of the role of white-grey matter interface
 - Finite element modeling of human spinal cord
 - Modeling of tumor mechanobiology and computational nanomedicine

Research guidance: **PhD - 11** (4 completed, 7 ongoing), **Master's – 20** (18 completed, 2 ongoing)

Independently developed and taught the following courses:

Biomicrofluidics (BM 6070(2 credit)/BM 6233(3 credit))

A course that focuses on the comprehensive design and development of chip scale microfluidic designs for bioanalytical applications – the course began in 2013 as a 3-credit course (BM 6233) and renamed as a 2-credit course (BM 6070) later in 2015. The course remains a favorite of final year undergraduate, masters and PhD students from various disciplines.

Advanced Biomechanics (BM 6080(2 credit)/BM 6230(3 credit))

Course which is focused on the principles of biomechanics applied to human body and the applications of linear and non-linear biomechanical models – the course began in 2013 as a 3-credit course (BM 6230) and renamed as a 2-credit course (BM 6080) later in 2015. The course remains a favorite of final year undergraduate, masters and PhD students from biomedical and mechanical engineering.

Lab on Chip (BM 5110(1 credit))

Introductory course on the principles of microfluidic chip scale devices and their applications in bioengineering – the course started from 2015 Aug, and has received wide appreciation from students of multiple engineering branches, as observed from yearly registration data.

Biomechanics (BM 5040(1 credit))

Basic course on biomechanics of human body aimed at BME masters students who are having interdisciplinary background to learn the importance of mechanics in bioengineering.

Introduction of Biomechanics (BM 3040(2 credit))

Third year undergraduate level introductory course on the mechanics of human body

Biomechanical Design (BM 4135(2 credit))

A recently floated project-based course on design of solutions to biomechanics problems – focused on third/final year undergraduate students

Co-development/contributed lectures:

Delivered lectures on Lab on Chip technology as part of BME department common course, 'Bioengineering' (BM1030)

TEACHING SUMMARY AT IIT HYDERABAD

(Feedback scores are indicated over maximum score of 5, **Average feedback over the courses taught during 2013-2025 = 4.55**). NA = Not available on AIMS, * - Data tracked through AIMS and ERP in 2024.

Sl No.	Course Number	Course Title	Year-Semester	Number of Credits	Number of students	Feedback obtained
1	BM 5040	Biomechanics	Aug-Nov 2025	1	33	4.82
2	BM 3040	Introduction to Biomechanics	Aug-Nov 2025	2	30	4.36
3	BM 6070	Biomicrofluidics	Jan-Apr 2025	2	36	NA
4	BM 6080	Advanced Biomechanics	Jan-Apr 2025	2	16	NA
5	BM 3040	Introduction to Biomechanics	Aug-Nov 2024	2	18	4.38*
6	BM 4135	Biomechanical Design	Aug-Nov 2024	2	9	4.97*
7	BM 5040	Biomechanics	Aug-Nov 2024	1	47	4.53*
8	BM 5110	Lab on Chip	Aug-Nov 2024	1	55	4.77*
9	BM 6070	Biomicrofluidics	Jan-Apr 2024	2	21	4.74*
10	BM 6080	Advanced Biomechanics	Jan-Apr 2024	2	6	4.4*
11	BM 5110	Lab on Chip	Aug-Nov 2023	1	29	4.34
12	BM 5040	Biomechanics	Aug-Nov 2023	1	13	4.69
13	BM 4135	Biomechanical Design	Aug-Nov 2023	2	4	4.35
14	BM 3040	Introduction to Biomechanics	Aug-Nov 2023	2	18	4.66

15	BM 6070	Biomicrofluidics	Jan-Apr 2023	2	16	4.34
16	BM 6080	Advanced Biomechanics	Jan-Apr 2023	2	8	4.32
17	BM 5110	Lab on Chip	Aug-Nov 2022	1	16	4
18	BM 5040	Biomechanics	Aug-Nov 2022	1	16	4.25
19	BM 3040	Introduction to Biomechanics	Aug-Nov 2022	2	16	4.29
20	BM 6070	Biomicrofluidics	Jan-Apr 2022	2	15	4.69
21	BM 6080	Advanced Biomechanics	Jan-Apr 2022	2	5	4.77
22	BM 5110	Lab on Chip	Aug-Nov 2021	1	23	4.38
23	BM 5040	Biomechanics	Aug-Nov 2021	1	42	4.31
24	BM 6070	Biomicrofluidics	Jan-Apr 2021	2	7	4.68
25	BM 6080	Advanced Biomechanics	Jan-Apr 2021	2	1	4.9
26	BM 5110	Lab on Chip	Aug-Nov 2020	1	17	4.7
27	BM 5040	Biomechanics	Aug-Nov 2020	1	15	4.87
28	BM 6070	Biomicrofluidics	Jan-Apr 2020	2	6	4.35
29	BM 6080	Advanced Biomechanics	Jan-Apr 2020	2	4	3.92
30	BM 5110	Lab on Chip	Aug-Nov 2019	1	11	4.47
31	BM 5040	Biomechanics	Aug-Nov 2019	1	10	4.4
32	BM 6070	Biomicrofluidics	Jan-Apr 2019	2	9	5
33	BM 6080	Advanced Biomechanics	Jan-Apr 2019	2	8	4.24
34	BM 5110	Lab on Chip	Aug-Nov 2018	1	20	4.65
35	BM 5040	Biomechanics	Aug-Nov 2018	1	17	4.28
36	BM 6070	Biomicrofluidics	Jan-Apr 2018	2	4	4.95
37	BM 6080	Advanced Biomechanics	Jan-Apr 2018	2	1	4.95
38	BM 5110	Lab on Chip	Aug-Nov 2017	1	12	4.9
39	BM 5040	Biomechanics	Aug-Nov 2017	1	14	4.5
40	BM 6070	Biomicrofluidics	Jan-Apr 2017	2	8	4.69
41	BM 6086	Seminars in Biomedical Engineering	Jan-Apr 2017	1	9	4.75
42	BM 6080	Advanced Biomechanics	Jan-Apr 2017	2	7	4.86
43	BM 5110	Lab on Chip	Aug-Nov 2016	1	17	4.75
44	BM 5040	Biomechanics	Aug-Nov 2016	1	19	No records
45	BM 6070	Biomicrofluidics	Jan-Apr 2016	2	13	4.13
46	BM 6086	Seminars in Biomedical Engineering	Jan-Apr 2016	1	11	4
47	BM 6080	Advanced Biomechanics	Jan-Apr 2016	2	4	4.3
48	BM 5110	Lab on Chip	Aug-Nov 2015	1	23	4.02
49	BM 5040	Biomechanics	Aug-Nov 2015	1	12	4.74
50	BM 6233	Biomicrofluidics	Jan-Apr 2015	3	4	4.7
51	BM 6086	Seminars in Biomedical Engineering and Healthcare	Jan-Apr 2015	2	10	3.65
52	BM 6230	Biomechanics	Aug-Nov 2014	3	12	4.78
53	BM 6233	Biomicrofluidics	Jan-Apr 2014	3	12	No records
54	BM 6086	Seminars in Biomedical Engineering and Healthcare	Jan-Apr 2014	2	9	No records
55	BM 6230	Biomechanics	Aug-Nov 2013	3	11	No records

PROJECTS

1. Neurological implications of traumatic brain injury – multiscale modeling of brain strain-induced Tau protein aggregation, SERB MATRICS (2023 – 2026) – 6.6 Lakhs (as PI)
2. Personalized precision oncology: 3D-printed microfluidic cancer-on-chip device for clinical validation of prior drug sensitivity using carcinoma breast cancer, ICMR (2024-2027), - 82.66 lakhs (as Co-PI)
3. Project titled "Rapid, affordable, portable SARS-CoV-2 screening kit for resource-limited settings, SERB (2020-2021) – 14.42 Lakhs (as Co-PI)
4. Novel growth-factor-nanoencapsulated 'functional-biomaterials' for the treatment of myocardial infarction, ICMR (2019-2022) – 1.28 crores (as Co-PI)

AWARDS AND HONORS

1. IITH Director's email of appreciation for teaching excellence in 2023
2. IES Engineering Awards, Singapore, 2006

PUBLICATIONS

Book Chapter

1. **Harikrishnan N. Unni** and Chun Yang, Deposition of colloidal particles from electrokinetic flow suspension in microfluidic channel, **Biofouling: Types, Impact and Anti-Fouling, Nova Science Publishers**, pp. 271-293 (2009).
2. Tanu Khanuja and **Harikrishnan Narayanan Unni**, Finite Element Analysis of Traumatic Brain Injury due to Blunt Impact of Different Durations, **Advances in Biomedical Engineering and Technology, Springer Nature** (2020), pp. 221-229.

Key Journals

1. Shobhit Das, **Harikrishnan Narayanan Unni**, Advancements in microfluidic droplet generation: methods and insights, **Microfluidics and Nanofluidics**, **29(4)**, **2025. (Impact factor: 2.8, Q2)**
2. Tony Thomas, Mano Govindharaj, **Harikrishnan Narayanan Unni**, Neema Kumari, Subha Narayan Rath, On-chip mixing of cancer cells and drug using LED enabled 2D opto-wetting

- droplet platforms, **Biomedical Physics and Engineering Express**, 9(4), 2023. (Impact factor: 1.4, Q3)
3. Maneesha Shaji, Mudigunda Sushma, Tejaswini Appidi, Shubha Jain, Aravind Kumar Rengan*, **Harikrishnan Narayanan Unni***, Microfluidic design of tumor vasculature and nanoparticle uptake by cancer cells, **Microfluidics and Nanofluidics**, 25 (46), 2021. (Impact factor: 2.8, Q2)
 4. Shubha Jain, Sarpras Swain, Lopamudra Das, Sarita Swain, Lopamudra Giri, Anand Kumar Kondapi, **Harikrishnan Narayanan Unni***, Microfluidic protein imaging platform: Study of tau protein aggregation and Alzheimer's drug response, **MDPI Bioengineering**, 7(4), 162, 2020. (Impact factor: 4.6, Q2)
 5. Shubha Jain, **Harikrishnan Narayanan Unni***, Numerical Modeling and Experimental Validation of Passive Microfluidic Mixer Designs for Biological Applications, **AIP Advances**, 10, 105116, 2020. (Impact factor: 1.6, Q2)
 6. Tanu Khanuja, **Harikrishnan Narayanan Unni***, Intracranial Pressure Based Validation and Analysis of Traumatic Brain Injury using a New Three Dimensional Finite Element Human Head Model, **I MechE: Journal of Engineering in Medicine**, 234 (1), 2020. (Impact factor: 1.76, Q3)
 7. Tony Thomas, **Harikrishnan Narayanan Unni***, LED-based optowetting and fluidic transport for droplet mixing, **Microfluidics and Nanofluidics**, 23 (105), 2019. (Impact factor: 2.8, Q2)
 8. Manjoosha R. Yerrapragada and **Harikrishnan Narayanan Unni***, Paper based Microfluidic Device for diagnosis of Osteoporosis markers, **Bioanalysis**, 10 (20), 2018. (Impact factor: 2.69, Q2)
 9. **Harikrishnan Narayanan Unni***, Deny Hartono*, Lin Yue Lanry Yung, Mary Mah-Lee Ng, Heow Pueh Lee, Boo Cheong Khoo and Kian Meng-Lim, Characterization and Separation of Cryptosporidium and Giardia Cells using On-chip Dielectrophoresis, **Biomicrofluidics**, 06, 012805 (2012). (The above article was selected by the **Virtual Journal of Biological Physics Research**, American Institute of Physics, Vol.23, Issue 6, 2012). (Impact factor: 3.26, Q2)
 10. **Harikrishnan Narayanan Unni***, Deny Hartono*, Kian-Meng Lim, On-Chip Trapping and Characterization of Cryptosporidium Using Surface Coated ITO-PDMS Bonded Chips, **Advanced Materials Research**, Vol. 254, pp. 191-194 (2011).
 11. **Unni HN**, Yang C., Colloidal particle deposition from electrokinetic flow in a microchannel, **Electrophoresis**, Vol. 30, 5, pp.732-741 (2009). (Impact factor: 3.53, Q2)

12. **H.N. Unni**, C. Yang, Kinetics of colloidal particle deposition to a solid surface from pressure driven microchannel flows, **Canadian Journal of Chemical Engineering**, Vol. 85, 5, pp 609-616 (2007). (**Impact factor: 2.5, Q2**)
13. **H.N. Unni**, H.J. Keh and C.Yang, Analysis of Electrokinetic transport in a microchannel, **Electrophoresis**, Vol. 28, pp 658-664 (2007). (**Impact factor: 3.53, Q2**)
14. **H.N.Unni** and C.Yang, Brownian dynamic simulation and experimental study of particle deposition in a microchannel flow. **Journal of Colloid and Interface Science**, Vol. 291, pp. 28-36 (2005). (**Impact factor: 9.9, Q1**)

Conferences

1. Tanu Khanuja, **Harikrishnan Narayanan Unni**, Computational Modeling of Translational Acceleration-Induced Diffuse Injury: The Role of Brain Inhomogeneity, 25th IEEE International Conference on Bioinformatics and Bioengineering (**IEEE BIBE 2025**), November 06-08, 2025, Greece (accepted).
2. Athul Vidya Rajeev, Varad Ashishrao Talnikar, **Harikrishnan Narayanan Unni**, Molecular dynamics simulation of TRPV1-mediated nanoparticle transport across normal and cancer cell membrane models, 25th IEEE International Conference on Bioinformatics and Bioengineering (**IEEE BIBE 2025**), November 06-08, 2025, Greece (accepted).
3. Vankudavath Rajunaik, **Narayanan Unni Harikrishnan**, Perugu Shyam, B Preethi and J Parasannanavar Devraj, An in silico approach to study Tauopathies in Alzheimer's disease and modeling of neuroprotective compounds targeting Neurofibrillary tangles, 25th IEEE International Conference on Bioinformatics and Bioengineering (**IEEE BIBE 2025**), November 06-08, 2025, Greece (accepted).
4. Athul Vidya Rajeev, Gaurav Kumar Singh, Himanshu Joshi, **Harikrishnan Narayanan Unni**, Molecular Modelling of Nanoparticle Delivery through Normal and Cancer Cell Membranes, Proceedings of the 10th World Congress on Electrical Engineering and Computer Systems and Sciences (EECSS'24) - **ICEBS 2024**, August 19-21 2024, Spain.
5. Sriram Hebbaluguppe Krishna, Viswanath Chinthapenta, Mohan Raghavan, **Harikrishnan Narayanan Unni**, Finite element modelling of an anatomically accurate human spinal cord, Proceedings of the 10th World Congress on Electrical Engineering and Computer Systems and Sciences (EECSS'24) - **ICEBS 2024**, August 19-21, 2024, Spain.
6. Shubha Jain, **Harikrishnan Narayanan Unni**, On-Chip platform for tau protein aggregation and Alzheimer's drug dose response, International Conference on Miniaturized Systems for Chemistry and Life Sciences (**MicroTAS**), 27-31 October 2019, Basel, Switzerland.
7. Tanu Khanuja, **Harikrishnan N. Unni**, Computational modelling of Traumatic Brain Injury

- due to impact on different sides of head, **IEEE Conference on Bioinformatics and Bioengineering (IEEE BIBE)**, October 29-31, 2018, Taiwan.
8. Shubha Jain, **Harikrishnan N. Unni**, Early-stage detection of Osteoporosis and markers using Lab on Chip mixing designs, **EMBL Microfluidics Conference**, 15-17 July 2018, Heidelberg.
 9. Tony Thomas, **Harikrishnan Narayanan Unni**, LED based Optowetting Platforms for Microfluidics, **SPIE BIOS**, 27 January – 1 February, 2018, USA.
 10. Manjoosha R. Yerrapragada, **Harikrishnan Narayanan Unni**, Paper based Microfluidic Platforms for Osteoporosis Diagnosis, International Conference on Miniaturized Systems for Chemistry and Life Sciences (**MicroTAS**), October 22-26, 2017, USA.
 11. Manjoosha R. Yerrapragada, **Harikrishnan Narayanan Unni**, Paper based analytic platforms for diagnosis of bone diseases, International Conference on Micro and Nanoengineering (**MNE**), September 18-22 2017, Portugal.
 12. Virendra J. Majarikar, **Harikrishnan N. Unni**, Modelling of Electrokinetic Mixing in Lab on Chip Microfluidic devices, International Conference on Microfluidics and Nanofluidics (**ICMN**), September 7-8 2017, Tokyo.
 13. Faruk Azam, Shubha Jain, **H.N.Unni**, Automated Electrowetting based Digital Microfluidic Chips for Next Generation Sequencing and Sample Preparation, **Advances in Next Generation Sequencing, SELECTBIO**, May 5-6 2015, India.
 14. V. V. Unnikuttan, **H.N.Unni**, Acoustophoretic Microfluidic Device for Concentration and Manipulation of Cells, **Advances in Microfluidics and Lab on a Chip, SELECTBIO (2015)**, India.
 15. Faruk Azam, Shubha Jain, **H.N.Unni**, Electrowetting based Digital Microfluidic Transport On Chip, **Advances in Microfluidics and Lab on a Chip (2015)**, India.
 16. Shubha Jain, Faruk Azam, **H.N.Unni**, Advanced Microfluidic Mixing Device for the study of Macromolecule Dynamics, **Advances in Next Generation Sequencing, SELECTBIO**, May 5-6 2015, India.
 17. Shubha Jain, Faruk Azam, **H.N.Unni**, An Effective Design and Simulation for Microfluidic Passive Mixing with Geometric Variation, **Advances in Microfluidics and Lab on a Chip, SELECTBIO (2015)**, India.
 18. Sk. Faruk Azam, **Harikrishnan Narayanan Unni**, Electrowetting and Droplet Transport in Digital Microfluidic Chips, **COMOSOL CONFERENCE**, November 13-14 2014, Bangalore, India.

19. **Harikrishnan Narayanan Unni**, Modelling of Dielectrophoretic Cell Concentration in a parallel plate microfluidic device, India-Japan Workshop on Biomolecular Electronics and Nanotechnology for Environment Preservation (**IJWBME**), Dec 13-15 2013, Delhi, India.
20. **Harikrishnan Narayanan Unni**, Deny Hartono*, Kian-Meng Lim, On-Chip Trapping and Characterization of Cryptosporidium Using Surface Coated ITO-PDMS Bonded Chips, **International Conference on Materials and Advanced Technologies (ICMAT)**, June 26 – July 01 2011, Singapore.
21. **Harikrishnan Narayanan Unni**, Deny Hartono*, Kian-Meng Lim, On-chip characterization of Cryptosporidium using Dielectrophoresis, **International Conference on Advances in Microfluidics and Nanofluidics (AMN)**, January 05–07 2011, Singapore.
22. **H. N. Unni and Chun Yang**, Kinetics of colloidal particle deposition from electrokinetic microfluidics flows, **7th International Conference on Nanochannels, Microchannels and Minichannels (ICNMM)**, June 22-24 2009, Pohang, South Korea.
23. Z. G. Li, L. K. Chin, H. J. Huang, **H. N. Unni** and A. Q. Liu, On-chip liquid tunable grating using laminar microfluidic control system, **12th International conference on miniaturized systems for chemistry and life sciences (MicroTAS)**, October 12-16 2008, San Diego, USA.
24. **H.N Unni** and C. Yang, "Deposition of colloidal particles from pressure driven microfluidic flow – Brownian Dynamics Simulation", **International Conference on Computational Methods**, December 15-17 2004, Singapore.

INVITED TALKS

1. **Microfluidics design towards brain-on-a-chip models**, 25 July 2024, IIT Bombay (Mechanical Engineering Department), India.
2. **Microfluidics design towards brain-on-a-chip models**, 16th International conference on sensing technology, 19 December 2023, BITS-PILANI Hyderabad, India.
3. **Research methods in bioengineering – Scopes and Perspectives**, Faculty development program, 02 July 2021, SSN College of Engineering, Chennai, India.
4. **Microfluidic approaches for circulating tumor cells – concentration, Identification and characterization**, SELECTBIO Microfluidics and Lab-on-chip conference, 12 September 2019, Mumbai, India

5. **Paper based microfluidic designs (muPAD) and applications**, SELECTBIO Microfluidics conference, 17 January 2018, Mumbai, India
6. **Micro-Nano Fluidics and Lab-on-a-chip technology**, Research Centre Imarat (RCI) - DRDO, 13 November 2013, Hyderabad, India.

REFERENCES

1. **Yang Chun, Charles**

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